Quant Modeling Project: Housing Price Prediction

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Course: AI & Python| Lei Ge | 2025 Fall

Final Innovations:

(1) **Optuna tuning (RF & XGBoost)**
→ systematic hyperparam search

(2) **NLP from customer feedback**
→ text → features

(3) **Explainability**
→ Feature importance + PDP/ICE sanity-check

Metrics	In sample rmse	out of sample rmse	Cross-validation rmse	Kaggle Score
Midterm Linear Model	367,166.44	349,180.33	353,852.85	54.7
Random Forest	66,097.27	147,587.70	155,210.82	74.3
ANN	195,883.17	232,562.11	240,287.81	71.0
Best Model (XGBoost)	65,055.23	139,462.93	145,361.22	75.4

Key Takeaways:

(1) **Kaggle +19.6** (54.7 → 74.3)

(2) **OOS RMSE ↓51.6%** (1.97M → 0.95M)

(3) **Best = Optuna-tuned XGBoost** (holdout & Kaggle consistent)

Metrics	In sample rmse	out of sample rmse	Cross-validation rmse	Kaggle Score
Midterm Linear Model	2,230,305.94	1,965,184.37	2,012,607.84	54.7
Random Forest	1,014,529.78	1,119,572.58	1,150,020.77	74.3
ANN	992,901.88	1,107,057.94	1,125,246.99	71.0
Best Model (XGBoost)	644,076.01	951,723.36	981,918.16	75.4

Note: RMSE on original price scale; train-only fit for feature pipeline; reproducible results (fixed seed).

#1. NLP (Rule-first + Local LLM)

Rule-first triage

↓

Local Gemma (JSON constrained)

↓

sentiment dummies

↓

modeling features

情感标签	
中性	38109
负面	32583
正面	28207

```
value counts:
fb_sent_正面     0.525325
fb_sent_负面     0.428221
dtype: float64
```

```
[sentiment] 总行数=103871 | 规则已判=62678 | 仍需API=41193
```

```
[info] rows=98899, nonempty=98899, unique_nonempty=98748
[cache] loaded=65142 from sent_cache.csv
[rule] labeled=0, need_llm=33606 (these will call API)
```

·确定性保证:
temperature=0, schema-forced JSON output

·稳健性保证:
retries + timeout + fallback-to-neutral

#2. XGBoost Winning Strategy

Optuna setup（系统搜参）

· Subsample 15,000 + 6-fold CV（seed=42，可复现）
· Native xgb.train( ) + early stopping（xgboost==3.1.2 稳定）
· TPE sampler + Median pruner（剪枝提速）

↓

Best params（最优超参）

eta=0.0668, max_depth=10, min_child_weight=5.455
subsample=0.785, colsample_bytree=0.877
gamma=0.0376, lambda(L2)=0.0750, alpha(L1)=0.1167
ecommended rounds ≈ 200,
tree_method: 'hist'ecommended

↓

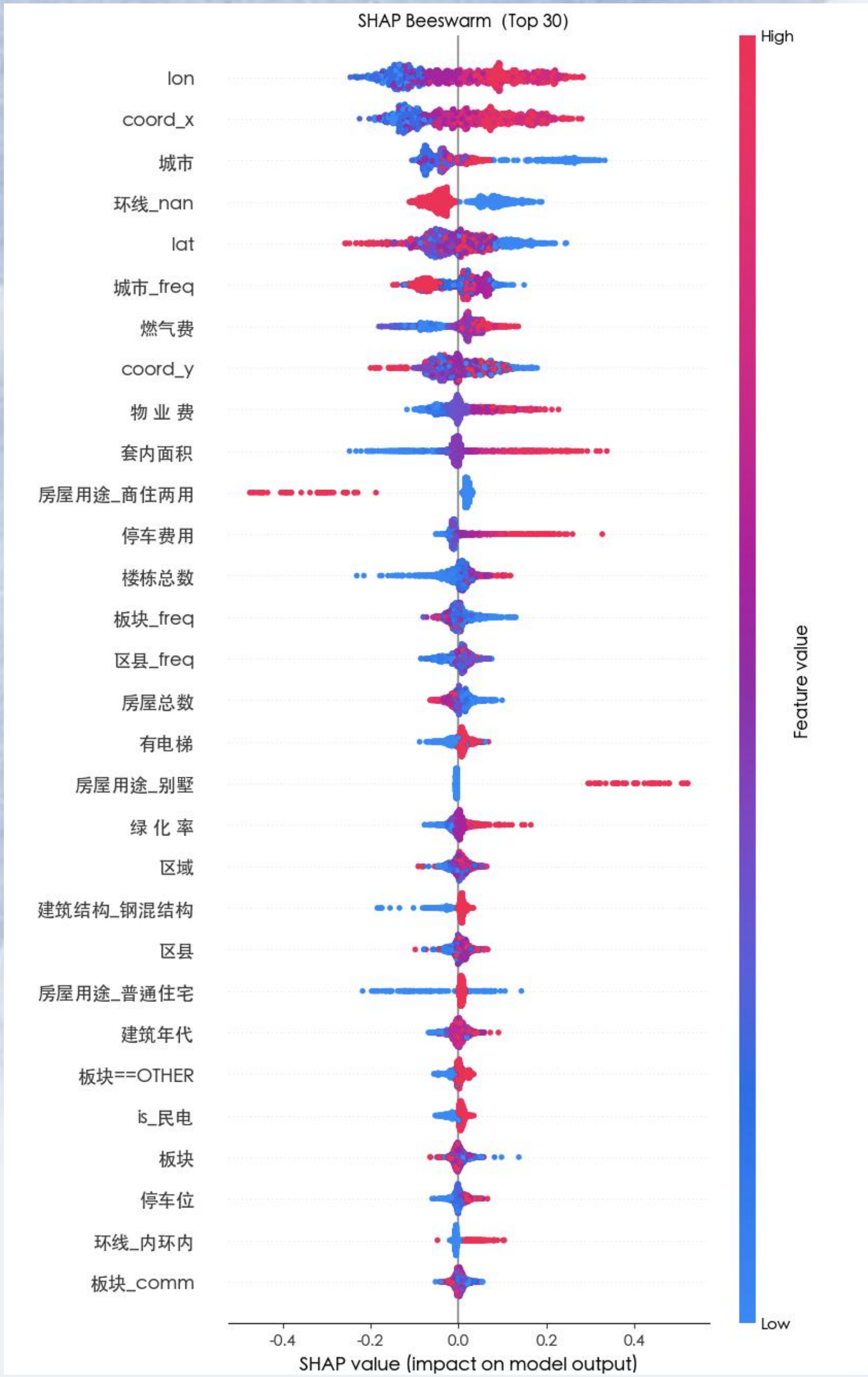
Why it wins（主要优势）

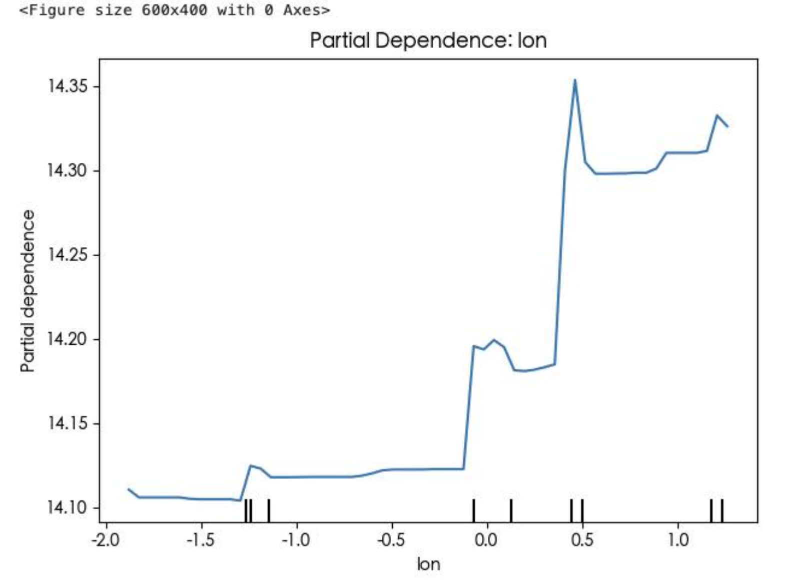
· **Deep tree + regularization**（L1/L2 + min_child_weight）
→ less overfit（更稳）

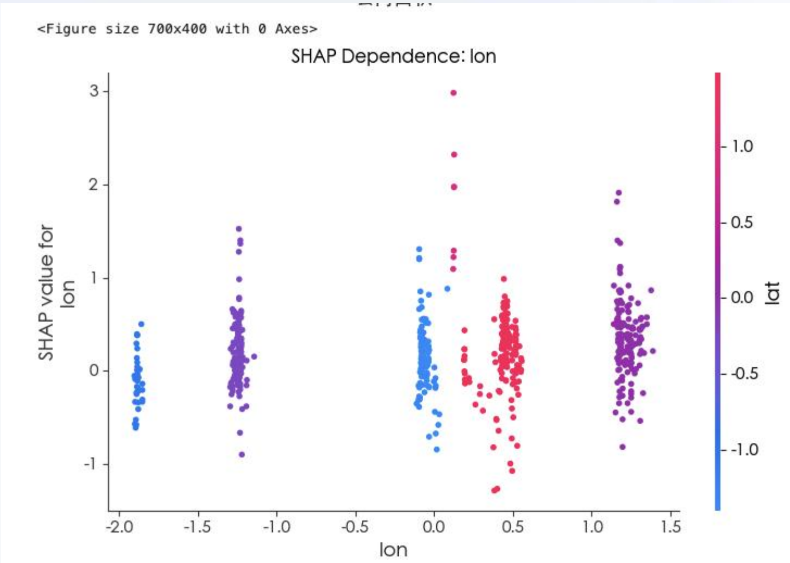
· **subsample/colsample 在0.7左右**
→ better generalization（降低方差）

· **captures nonlinear + interactions**
（含 NLP sentiment dummies）

Note: tuned on log-target for stability；final metrics reported on original scale







Explainability

（1）Location dominates — lon/lat/coord + city are top drivers
（地理位置是房价核心驱动，符合常识）

（2）Nonlinear / stepwise
“跨过某些地理分区/城市边界，价格水平整体换档”

（3）Interaction: longitude effect depends on latitude
（交互：经度效应受纬度调制）
东西 × 南北共同决定溢价）